

Genome-wide association analyses in East Asians identify new susceptibility loci for colorectal cancer

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To identify new genetic factors for colorectal cancer (CRC), we conducted a genome-wide association study in east Asians. By analyzing genome-wide data in 2,098 cases and 5,749 controls, we selected 64 promising SNPs for replication in an independent set of samples, including up to 5,358 cases and 5,922 controls. We identified four SNPs with association P values of 8.58×10^{-7} to 3.77×10^{-10} in the combined analysis of all east Asian samples. Three of the four were replicated in a study conducted in 26,060 individuals of European descent, with combined P values of 1.22×10^{-10} for rs647161 (5q31.1), 6.64×10^{-9} for rs2423279 (20p12.3) and 3.06×10^{-8} for rs10774214 (12p13.32 near the CCND2 gene), derived from meta-analysis of data from both east Asian and European-ancestry populations. This study identified three new CRC susceptibility loci and provides additional insight into the genetics and biology of CRC.